

Project: DW/Big Data Implementation (20%)

Coexist between Traditional DE & Modern DE (Big Data)

2190436 Data Warehousing

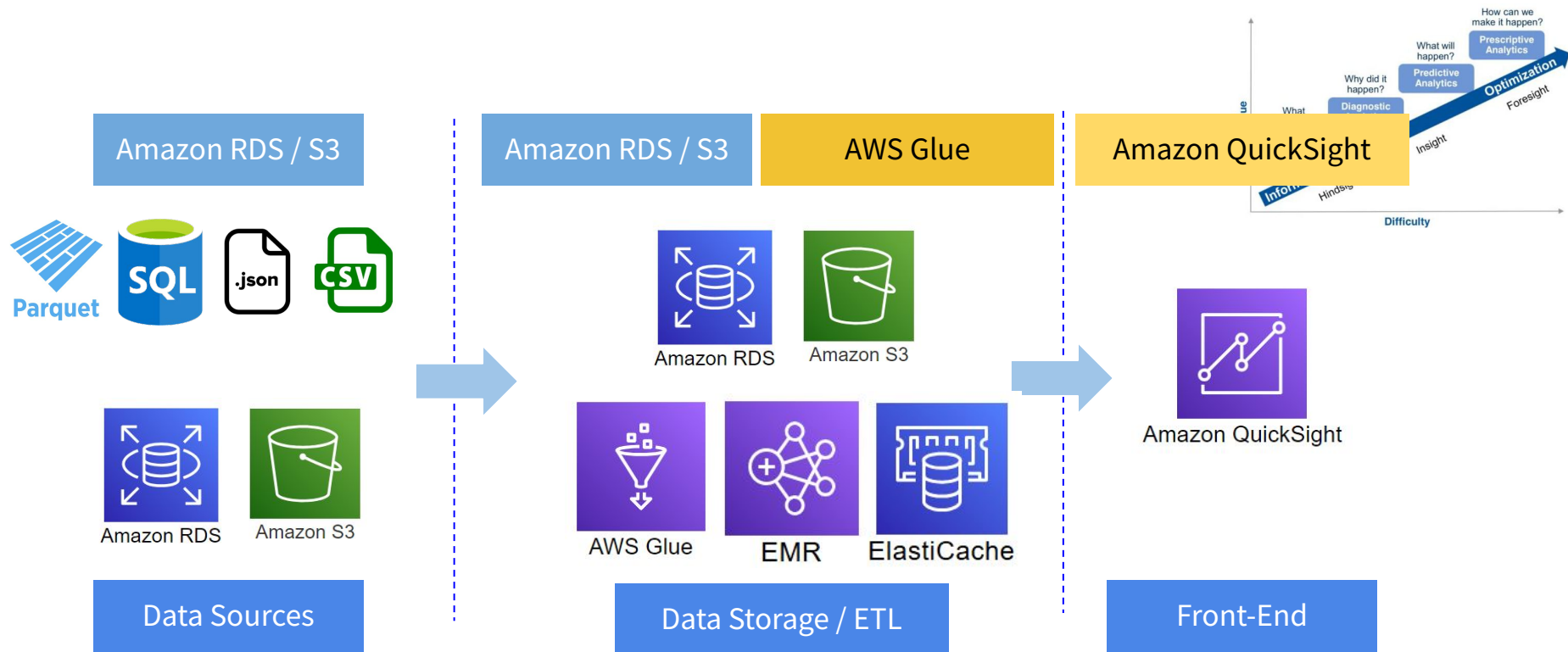
2190518 Big Data & Data Engineering

17 Mar 2026 (2025/2)



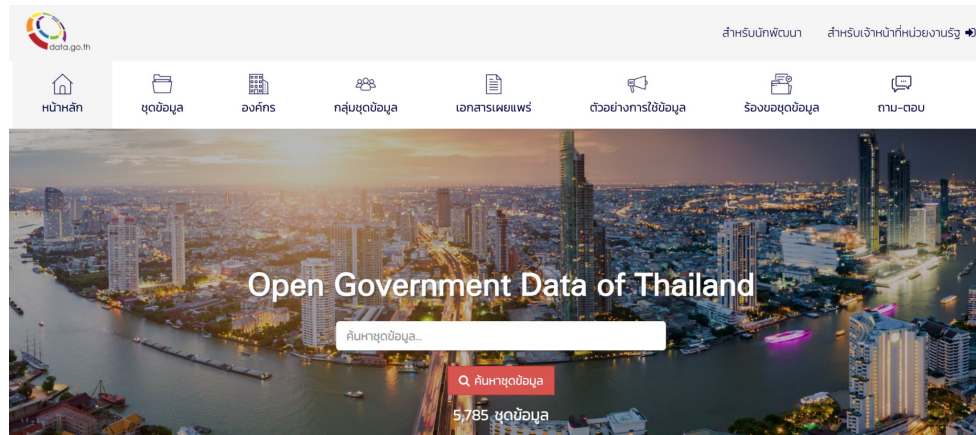
Objective: Full DW pipeline on AWS + Big Data

Coexist between Traditional DE & Modern DE (Big Data)



You can choose your own data source

- It can be public or private data.
- For example,
 - COVID-19
 - PM2.5
 - Scrape data from web by yourselves
 - <https://data.go.th/>
 - <https://www.kaggle.com/datasets>
 - <https://data.world/>



Score overview (Total 20%)

1.	Data source	2%
2.	DW	4%
3.	Big Data	4%
4.	Automate Workflow	1%
5.	BI	4%
6.	Presentation 2 videos	5% (2.5%*2)

Project Details

1) Data Collection & Source DB [2%]

- Load data into a source database.
- Use interesting & realistic datasets.
- Prefer multiple data sources (not just one CSV).
- Data integration / joins are encouraged.

2) Data Warehouse & ETL [4%]

- Design DW using 4-step dimensional modeling.
- Implement ETL with Amazon Glue.
- Include ≥ 1 [fact table](#) with conformed bus matrix (no aggregate fact).
- All dimension tables must implement SCD.
- Design correctness and DW techniques will be evaluated.

Project Details (cont.)

3) Big Data Component [4%]: Include one Big Data technology with justification, e.g.

- Spark → large data (Volume)
- Data Lake → multiple formats (Variety)
- KV database → fast lookup, infrequent updates

4) Automatic Workflow [1%]: Schedule pipelines using Glue Scheduler and/or Airflow.

5) BI Dashboard [4%]

- Create ≥ 2 dashboards (e.g., Overview → Detail).
- Implement using Amazon QuickSight.

Project Details (cont.)

6) Presentation [5%] (2 videos)

- **Video 1 (≤15 min) – Data Warehouse Designer/Data Engineer**
- Data preparation, DW design, ETL architecture, and demo Glue ETL running.
- **Video 2 (≤10 min) – Business Analyst**
- DW overview, QuickSight dashboards, key business insights.

More Details

- Maximum 4 students per group (can be 1, 2, 3 students per group, **CANNOT be 5!**)
- By **Fri 8th May 2026**, the following items must be submitted on MyCourseVille [here](#).
 - Google Drive Link to all your resources (BI project, overall diagram, data source files, etc.)
 - Set sharing to Anyone with link
 - Youtube Video Link for Video 1 (15 mins)
 - Set visibility to Public
 - Youtube Video Link for Video 2 (10 mins)
 - Set visibility to Public

FAQ

Q1: Do we need to use the provided AWS services?

A: Yes. You should use the AWS services provided for the course because the instructors (admins) can access your assigned AWS project account to review and grade your work.

Q2: For DW [4%] and Big Data [4%], what is the main evaluation criterion?

A: The score will be based on (1) effort and (2) techniques used. For example, a very simple workflow (e.g., Data Source → Change Schema → Target Data) would receive fewer points compared to a more complex and well-designed pipeline with multiple processing steps. More sophisticated implementations will receive higher scores.